



Artificial Intelligence Foundations (CIS4049-N)

Week 2 – Intelligent Agents – Part 1

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Today's plan

- **Intelligent/rational agents**
- **Designing rational agents**
- **Environments of rational agents**



Intelligent/rational Agents

Artificial Intelligence

The science of making machines that:

Humanly vs. Rationally; Thinking vs. Acting

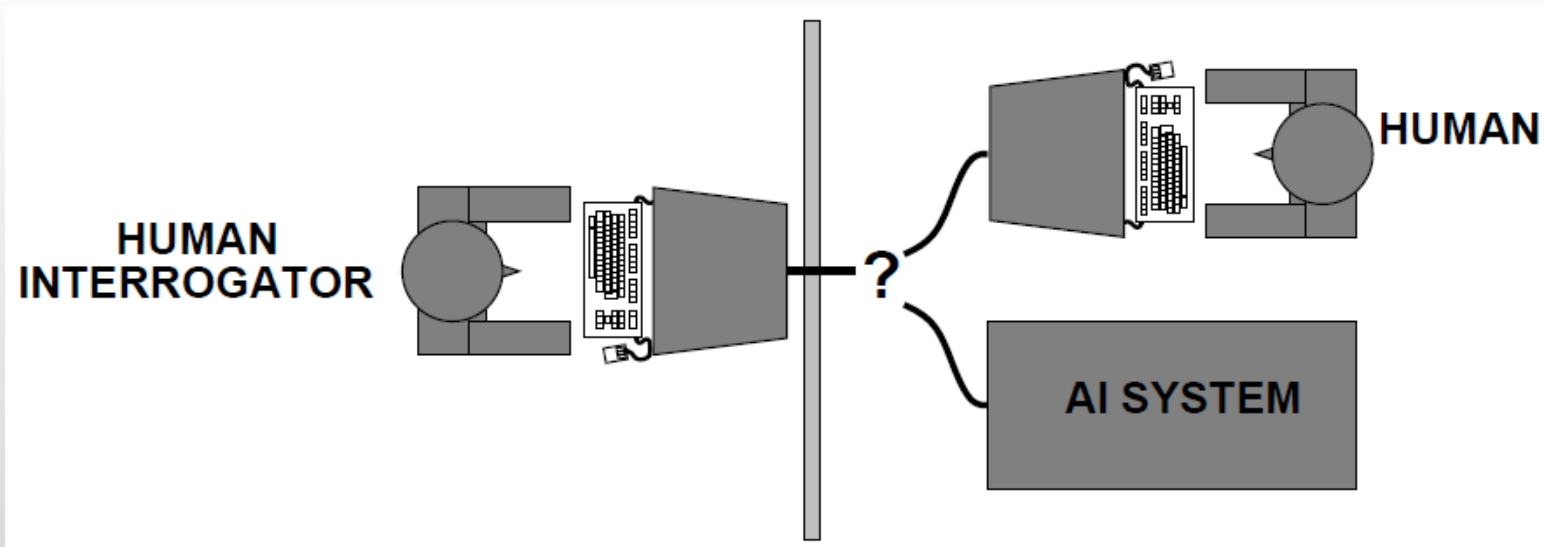
“[the automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning” (Bellman, 1978)	“the study of mental faculties through the use of computational models” (Charniak & McDermott, 1985)
“the study of how to make computers do things at which, at the moment, people are better” (Rich & Knight, 1991)	“the branch of computer science that is concerned with the automation of intelligent behavior” (Luger & Stubblefield, 1993)

Four typical categories:

thinking humanly	thinking rationally
acting humanly	acting rationally

Acting Humanly – Turing Test

- **Alan Turing**, Computing Machinery and Intelligence (**1950**): from “can machines think?” to “can machines act intelligently?”
- Operationalization: the **imitation game**

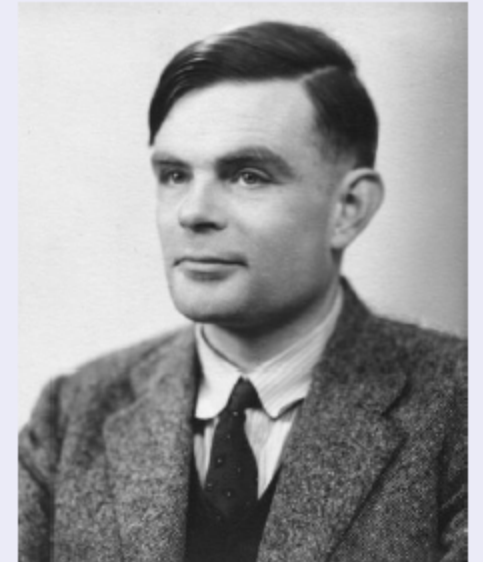


<http://www.imdb.com/title/tt2084970/>

Can machines think?

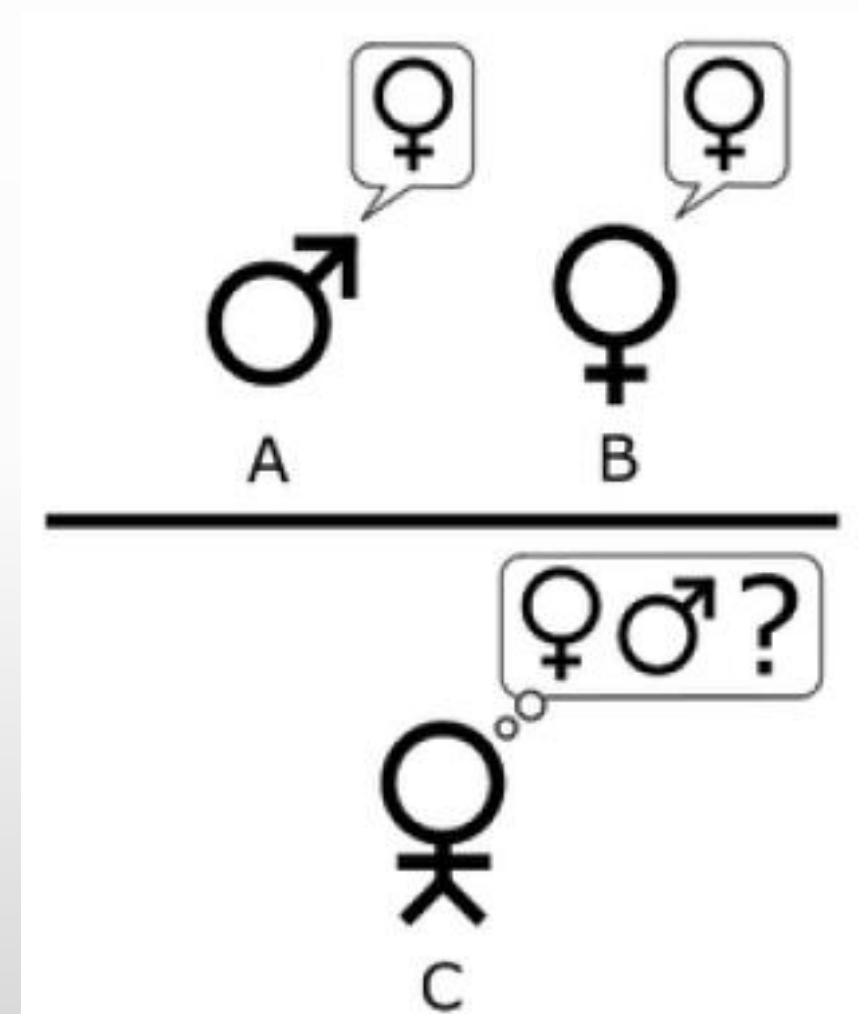
Alan Turing (1950)

*"I believe that **in about fifty years' time** it will be possible to program computers to make them play the imitation game so well that an average interrogator will not have more than 70 per cent chance of making the right identification after five minutes of questioning."*



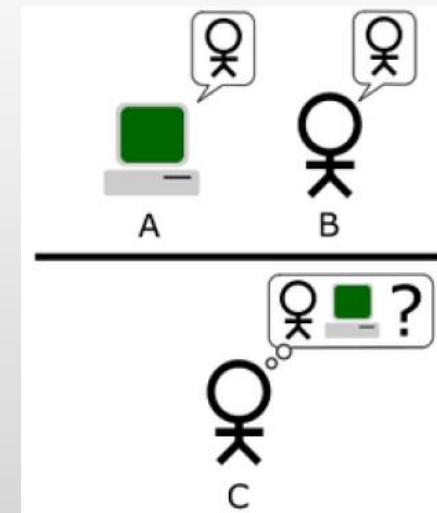
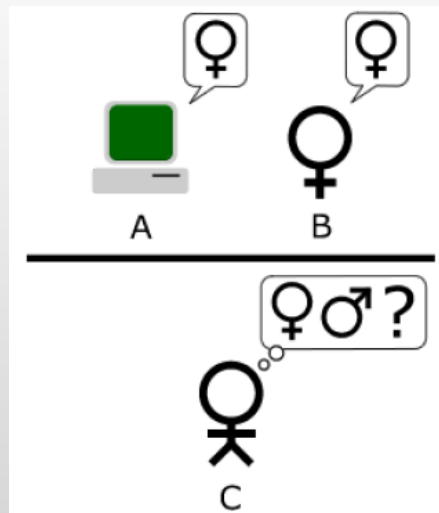
Turing Test

- The game includes three players with **different goals**: A, B (who answer questions) and C (who asks questions). A and B are of opposite sex
- **The goal of player C**: determine the sex of A and B by asking them questions
- **The goal of player B**: to help player C in his task
- **The goal of player A**: to trick player C into failing in his task
- The experiment is repeated several times measuring the success rate of player C



Turing Test

- What will happen if a machine assumes the role of player A?
- Will the question asker C make the same number of mistakes as in the case when both A and B are human?
- **Turing: if the number of mistakes is equal, then the machine is intelligent**
- Standard variant of the test: the question asker C must identify the human player



Acting Humanly – Turing Test

- What abilities would a machine need to have to pass the Turing Test?
 - **Natural language processing**
 - **Knowledge representation**
 - **Automated reasoning**
 - **Machine learning**
- Turing predicted that by the year 2000 computers (with about 120MB of memory) will have a 30% chance to fool humans

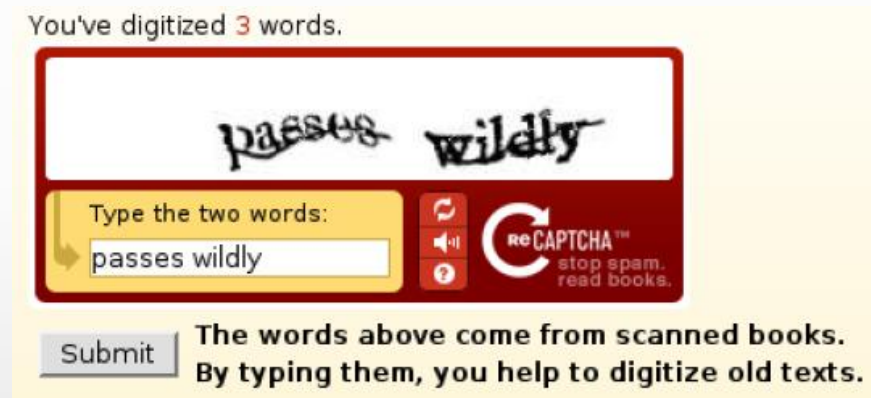
How useful is the Turing Test?

- Scientific usefulness is questionable
- Not important in AI “mainstream”
- Annual competitions (Loebner Prize): <https://en.wikipedia.org/wiki/LoebnerPrize>
- Practical application: CAPTCHA (“Completely Automated Public Turing test to tell Computers and Humans Apart”)



Reverse Turing test: CAPTCHA

- **CAPTCHA - Completely Automated Public Turing Test to Tell Computers and Humans Apart**



- A study (conducted using Amazon Mechanical Turk) shows that CAPTCHA is “often more complex than it should be” - the average solve time is 9.8 seconds (Bursztein et al., 2010)

Drawbacks of Turing Test

- Human vs. general intelligence
- Real vs. simulated intelligence
- Naivety of the question asker (proven in the case of ELIZA-bot:
<https://www.masswerk.at/elizabot/eliza.html>)

Irrelevance of the test

Aeronautics textbooks don't define aeronautics as:
"Building machines that fly so similarly to pigeons that they can fool other pigeons."



Turing's "Computing Machinery and Intelligence"

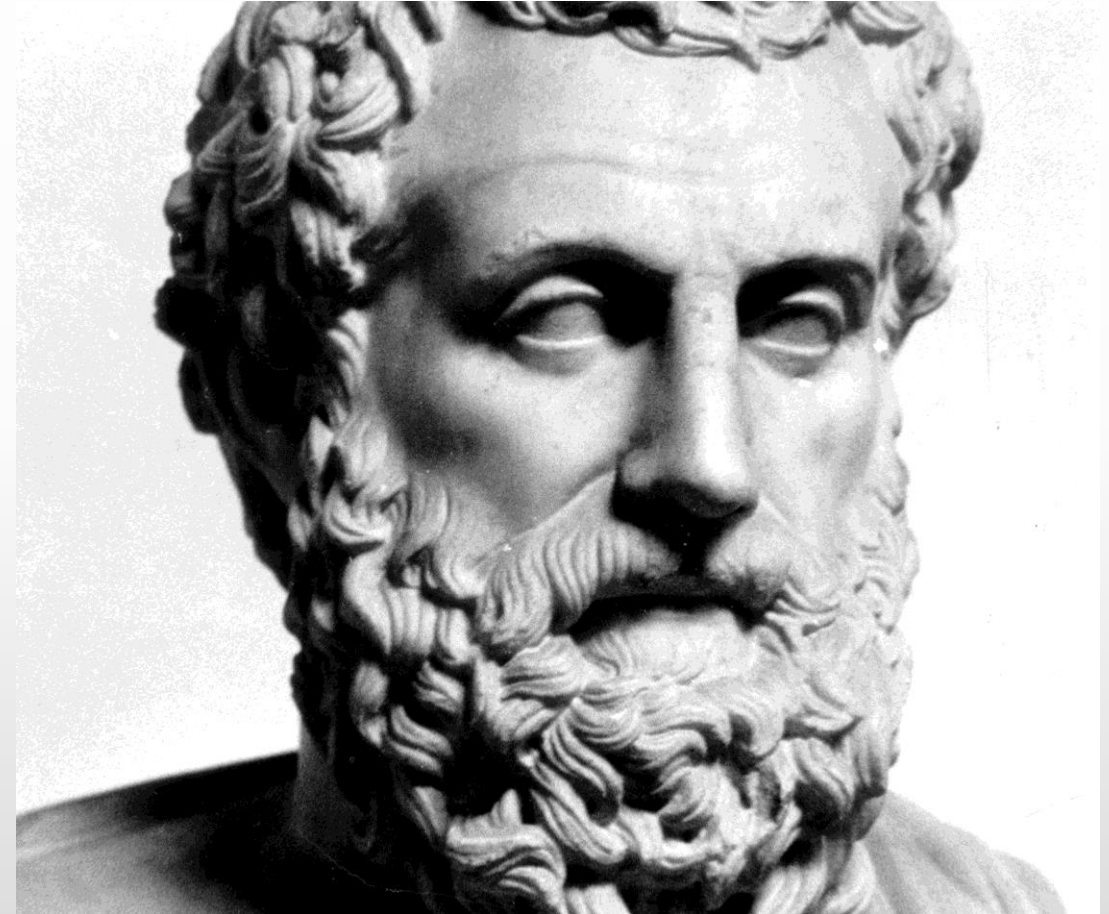
- Already discussed all important arguments of the 20th century against possibility of AI
- Suggested core aspects of AI: **knowledge representation, reasoning, language understanding, learning**
- **TT is perhaps more significant for the philosophy than for the development of AI**

Thinking Humanly - Cognitive Science

- **Cognitive revolution** of the 1960s: information processing supplants dominant behaviorism in psychology
- Which **cognitive abilities** are necessary for intelligent behavior? It requires scientific theory of brain activity
- Which level of abstraction? “knowledge” or “circuits”?
- How to test? Requires prediction/test of human behavior (top-down) or identification from neurological data (bottom-up)
- This roughly corresponds to **cognitive science** and **cognitive neuroscience**
 - today separate research areas from AI

Thinking Rationally – Laws of Thought

- **Normative (prescriptive)** rather than **descriptive**
- Aristotle: What are correct arguments/modes of thought?
- Syllogisms: structures for arguments that always yield correct conclusions given correct premises:
*“Socrates is a human. All humans are mortal.
Therefore Socrates is mortal.”*
- Several Greek schools of thought developed various forms of **logic**:
 - **Notations** (syntax) and **derivation rules** (calculi) for “correct” thinking
 - Direct connection to modern AI via mathematical logic (early 20th century)



Problems of the logical approach

- Not all intelligent behavior stems from **logical thinking**
- Which conclusions are **relevant**?
- How to deal with **uncertainty**?
- How to deal with **contradictions**?